

The TOP 5 IoT HACKS



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The Internet of Things (IoT) automates your lives by connecting everything around you from your cars to your homes to even your bodies.

On the other side of the spectrum is the concern for the safety and security of your data. All data is interconnected, which reduces the efforts needed by hackers to hack the entire system.

A large number of Internet-connected devices lack even the most basic cyber security protocols, making these hackable in minutes. And even though most secure and trusted machines give hackers a tough time, one tiny loophole is sufficient for them to steal your data. In fact, Intel's prediction for threats in 2017 includes IoT malware that will open backdoors into the connected home that could go undetected for years!

To add to the fear, there are a lot of such IoT hacks that have been reported in the recent past. Let us take a look at five of these.

You are on the driving seat, but it is not you who is driving

Charlie Miller and Chris Valasek, automotive cyber security researchers, have proved that hacking Jeeps is child's play for them, especially because all carmakers are doing their best to turn an automobile into a smartphone. Uconnect, an Internet-connected computer feature in hundreds of thousands of Fiat Chrysler cars, sport utility vehicles (SUVs) and trucks, controls the vehicles' entertainment and navigation, enables phone calls and offers Wi-Fi hot spots. Uconnect's mobile connection lets anyone who knows the car's

Internet Protocol (IP) address gain access from anywhere in the world.

In 2015, the duo attacked a Jeep over the Internet, from a distance of around 15 kilometres. They toyed with the air-conditioning, radio and windshield wipers, disabled brakes

at low speeds and managed to paralyse the vehicle. This made Chrysler announce for a recall of 1.4 million vehicles and fix the vulnerabilities. But now these researchers have come out with better hacking tricks.

By sending carefully-crafted messages on the vehicle's internal network known as a controller area network (CAN) bus, they can pull off even more dangerous, unprecedented tricks like causing unintended acceleration and slamming on the car's brakes, or turning the vehicle's steering wheel at any speed. Unlike last year, instead of cutting the transmission on the highway, they can turn the wheel 180 degrees.

One major relief is that this hack can be performed only with a laptop that is directly plugged into the Jeep's CAN network via a port under its dashboard. But they promise this would get wireless soon.

Instead of merely compromising one of the electronic control units (ECUs) on a target car's CAN network and using it to spoof messages to the car's steering or brakes, they attacked the ECU that sends legitimate commands to those components that would otherwise contradict their malicious commands and prevent attacks.

By putting the second ECU into bootrom mode—the first step in updating the ECU's firmware that a mechanic might use to fix a bug—hackers were able to paralyse that innocent ECU and send malicious commands to the target component without interference.

Adventures of Barnaby Jack

Barnaby Jack, a hacker programmer and computer security expert, has managed to hack multiple devices including automatic teller machines (ATMs), pacemakers, ear implants and insulin pumps. At Black Hat conference in 2010, he gave a presentation on jackpotting, or exploiting ATMs to make these dispense cash without using a bank's debit or credit card. Jack gave demonstrations about different kinds of attacks involving physical access to the machines and completely automated remote attacks. In both cases, malware was injected into

